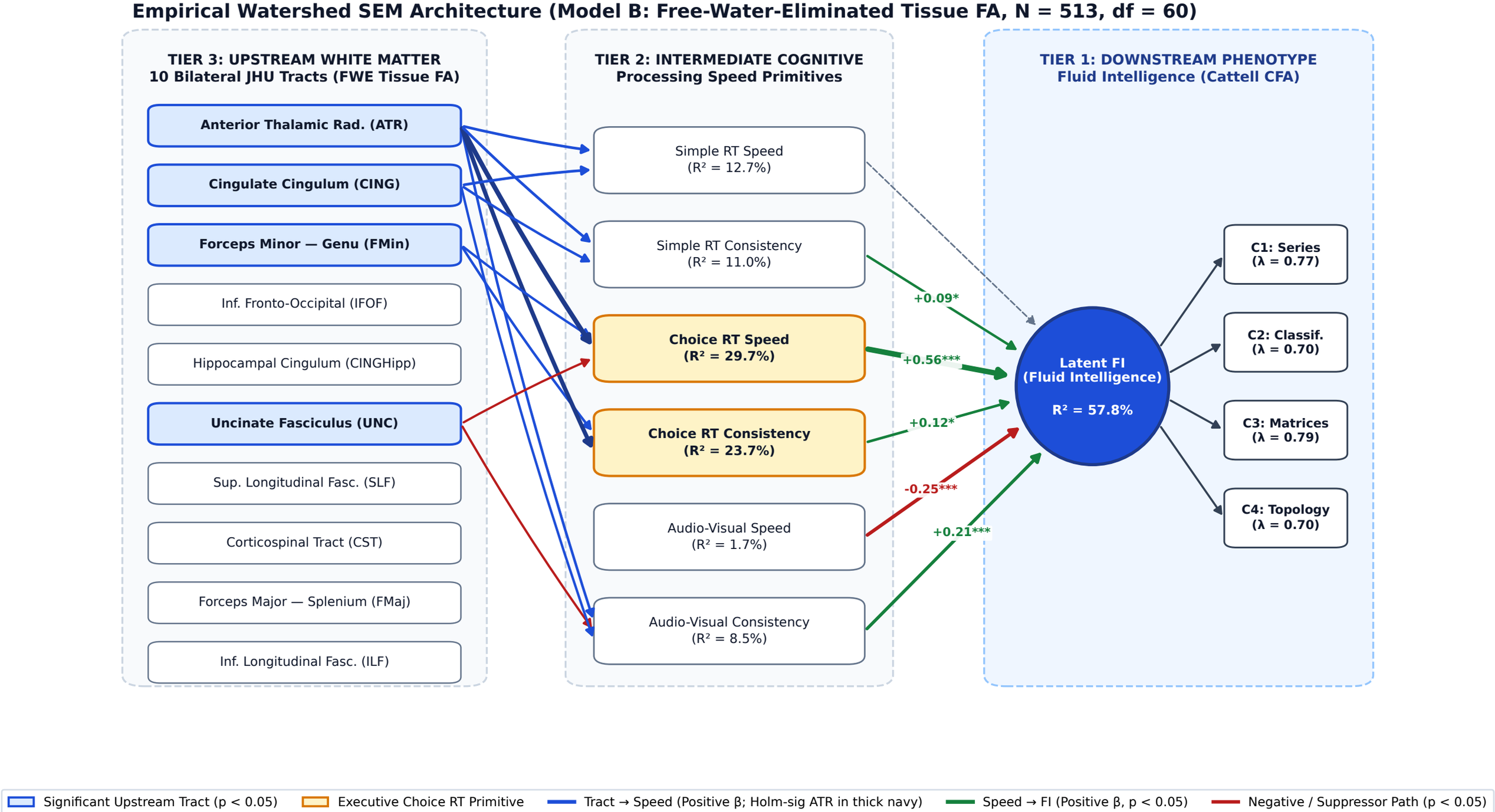
Empirical Watershed SEM Architecture (Model B: Free-Water-Eliminated Tissue FA, N = 513, df = 60)
TIER 3: UPSTREAM WHITE MATTER
10 Bilateral JHU Tracts (FWE Tissue FA)
Anterior Thalamic Rad. (ATR)
Cingulate Cingulum (CING)
Forceps Minor — Genu (FMin)
Inf. Fronto-Occipital (IFOF)
Hippocampal Cingulum (CINGHipp)
Uncinate Fasciculus (UNC)
Sup. Longitudinal Fasc. (SLF)
Corticospinal Tract (CST)
Forceps Major — Splenium (FMaj)
Inf. Longitudinal Fasc. (ILF)
TIER 2: INTERMEDIATE COGNITIVE
Processing Speed Primitives
Simple RT Speed
(R² = 12.7%)
Simple RT Consistency
(R² = 11.0%)
Choice RT Speed
(R² = 29.7%)
Choice RT Consistency
(R² = 23.7%)
Audio-Visual Speed
(R² = 1.7%)
Audio-Visual Consistency
(R² = 8.5%)
TIER 1: DOWNSTREAM PHENOTYPE
Fluid Intelligence (Cattell CFA)
Latent FI
(Fluid Intelligence)
R² = 57.8%
C1: Series
(λ = 0.77)
C2: Classif.
(λ = 0.70)
C3: Matrices
(λ = 0.79)
C4: Topology
(λ = 0.70)
+0.09*
+0.56***
+0.12*
-0.25***
+0.21***
Significant Upstream Tract (p < 0.05)
Executive Choice RT Primitive
Tract → Speed (Positive β; Holm-sig ATR in thick navy)
Speed → FI (Positive β, p < 0.05)
Negative / Suppressor Path (p < 0.05)